Instructions: Show all work in the space provided. Write each final answer in its box. Only what is inside a box is graded. Write the letter of the single best answer in its Letter box. Write intervals in interval notation. No notes. **70 points total.**

Part A — Rates of Change and Characteristics (14 pts)

Problems 1 and 2 use this table of values for a polynomial f.

x	0	1	2	3	4
$f(x)$	1	-2	1	16	49

1. Use successive differences to find the degree of the polynomial that fits f.

Degree: ☐

2. Find the average rate of change of f on the interval $[1, 4]$.

☐

3. Let $g(x) = x^2 - 3x$. Find and simplify the difference quotient $\dfrac{g(x+h) - g(x)}{h}$, $h \neq 0$.

☐

4. The graph of the polynomial f continues in the direction of its arrows. Its turning points are marked.

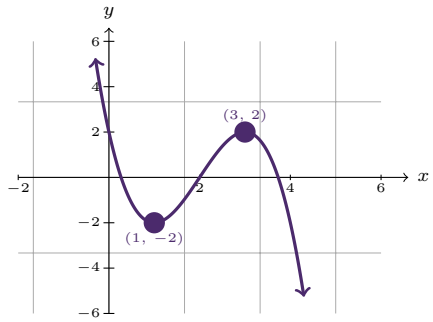
(a) The y-intercept: ☐

(b) The relative maximum value: ☐

(c) Increasing on: ☐

(d) Decreasing on: ☐

(e) The point of inflection: ☐



(f) Which is true?
(A) The absolute maximum of f is 2.
(B) The absolute maximum of f is -2.
(C) f has no absolute maximum.

Letter: ☐

Part B — Polynomial Behavior (12 pts)

5. Let $f(x) = 3x^2 - 2x^5 + x - 4$.

Degree	Leading coefficient	Most turning points	Most real zeros

6. Find the leading term of $g(x) = (1 - 2x)(x + 3)^3$.

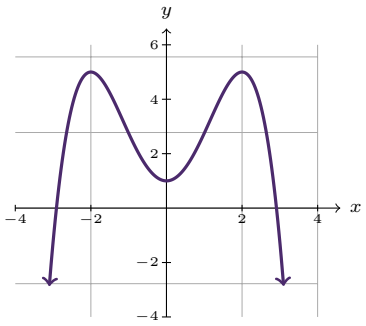
Leading term: ____

7. Give the end behavior of the same g in limit notation.

$\lim_{x \to -\infty} g(x) =$ ____

$\lim_{x \to \infty} g(x) =$ ____

8. The graph of the polynomial p is shown. It continues in the direction of its arrows.



(a) The least possible degree of p: ____

(b) Which describes the degree and the leading coefficient of p?

(A) even, positive (B) even, negative

(C) odd, positive (D) odd, negative

Letter: ____

Part C — Zeros and Multiplicity (10 pts)

9. Let $h(x) = (x + 3)^2(x - 1)(x - 4)^3$. Complete the table. Write *crosses* or *touches* for what the graph does at each zero.

Zero of h	-3	1	4
Multiplicity			
Crosses or touches?			

10. For the same h, is $h(x)$ *positive* or *negative* just to the left of $x = -3$? Just to the right of $x = -3$?

Just left: ____ Just right: ____

11. A polynomial of degree 3 with real coefficients has zeros 4 and $1 + 2i$. What is its third zero?

____

Part D — Factoring and Solving (11 pts)

For 12–14, factor completely.

12. $2x^3 - 18x$

13. $x^3 + 3x^2 - 4x - 12$

14. $x^3 + 64$

15. Solve $x^4 - 5x^2 - 36 = 0$. Give every solution, real and non-real.

$x =$

16. To solve $x^3 = 16x$, Dana divides both sides by x and gets $x^2 = 16$, so $x = \pm 4$. Which statement is true?

(A) Dana is right: the solutions are 4 and -4.

(B) Dana lost a solution: the solutions are 0, 4, and -4.

(C) Dana lost a solution: the solutions are 16, 4, and -4.

(D) Dana gained a solution: the only solution is 4.

Letter:

Part E — Dividing Polynomials (14 pts)

17. Use long division to divide $2x^3 + x^2 - 4x + 3$ by $x^2 + 1$. Fill in the boxes.

$2x^3 + x^2 - 4x + 3 = (x^2 + 1)(\quad\quad) + \quad\quad$

18. Use synthetic division to divide $2x^3 - 5x + 7$ by $x + 2$.

Quotient:

Remainder:

19. Use the Remainder Theorem to find the remainder when $p(x) = x^4 - 3x^3 + 2x - 5$ is divided by $x - 2$.

Remainder:

20. $x = 2$ is a zero of $p(x) = x^3 - 3x^2 + 4$. In the box: divide $p(x)$ by $x - 2$, write $p(x)$ factored completely, and list every zero of p with its multiplicity.

Part F — Writing Polynomials from Zeros (9 pts)

21. Write the polynomial with zeros -1, 2, and 3 and leading coefficient 1, in standard form.

22. A polynomial f of degree 3 has zero 2 with multiplicity 2, zero -1, and passes through $(0, 8)$. Find the leading coefficient a, then write $f(x)$ in factored form.

$a =$ $\qquad$ $f(x) =$

23. Write the polynomial of least degree with real coefficients, leading coefficient 1, and zeros 3 and $2i$, in standard form.